CO1_AT3

Question 1: Education Analytics Dataset

1. Python program

```
import pandas as pdimport matplotlib.pyplot as pltimport seaborn as snsfrom io import StringIOdata = """Student_ID,Attendance,Study_Hours,Internal_Marks,ResultS01,92,5,45,PassS02,55,2,24,FailS03,78,4,38,PassS04,60,3,28,FailS05,88,5,42,PassS06,47,1,20,FailS07,95,6,48,PassS08,72,3,35,PassS09,50,2,22,FailS10,83,4,40,PassS11,66,3,30,FailS12,90,5,44,PassS13,58,2,25,FailS14,76,4,36,PassS15,62,3,29,FailS16,85,5,41,PassS17,45,1,18,FailS18,80,4,39,PassS19,68,3,32,PassS20,53,2,23,Fail"""df = pd.read_csv(StringIO(data))print("First five rows:")print(df.head())print("\nRows and columns:", df.shape)print("\nData types:")print(df.dtypes)
```

2. Observations

1. The dataset contains 20 students and 5 columns.
2. There are 12 Pass students and 8 Fail students.
3. Students who study more hours generally obtain higher internal marks.
4. Students with higher attendance generally perform better.
5. No missing values or duplicate rows are present.

3. Conclusion

Attendance, study hours, and internal marks are important factors related to student performance. Regular attendance and consistent study may help students achieve better results.

Question 2: Healthcare Analytics Dataset

1. Python program

```
import pandas as pdimport matplotlib.pyplot as pltimport seaborn as snsfrom io import StringIOdata = """Patient_ID,Age,Sugar_Level,BP,BMI,Disease_StatusP01,45,145,130,28,YesP02,32,98,118,23,NoP03,55,180,145,31,YesP04,28,90,110,22,NoP05,60,200,150,34,YesP06,40,120,125,26,NoP07,52,165,140,30,YesP08,35,105,120,24,NoP09,48,155,135,29,YesP10,30,95,115,23,NoP11,62,210,155,35,YesP12,38,115,122,25,NoP13,50,170,142,30,YesP14,27,88,108,21,NoP15,58,190,148,33,YesP16,42,130,128,27,NoP17,46,150,132,28,YesP18,34,100,116,24,NoP19,53,175,144,31,YesP20,29,92,112,22,No"""df = pd.read_csv(StringIO(data))print("First five rows:\n", df.head())print("\nRows and columns:", df.shape)print("\nData types:\n", df.dtypes)print("\nMissing values:\n", df.isnull().sum())print("\nDuplicate rows:", df.duplicated().sum())print("\nStatistical summary:\n", df.describe())
```

2. Observations

1. The dataset contains 20 patients and 6 columns.
2. There are 10 patients with disease and 10 without disease.
3. Patients marked “Yes” generally have higher blood sugar levels.
4. Patients marked “Yes” generally have higher BMI and blood pressure.
5. The dataset has no missing values or duplicate rows.

3. Conclusion

Blood sugar, BMI, age, and blood pressure show patterns associated with disease status in this dataset. These observations are useful for preliminary analysis, but they cannot independently diagnose a disease or establish its cause.

Question 3: Retail Sales Dataset

1. Python program

import pandas as pdimport matplotlib.pyplot as pltimport seaborn as snsfrom io import StringIOdata = """Product_ID,Category,Price,Quantity_Sold,Discount,RevenuePR01,Grocery,50,20,5,1000PR02,Snacks,30,35,3,1050PR03,Dairy,45,18,2,810PR04,Fruits,60,25,5,1500PR05,Vegetables,40,30,4,1200PR06,Grocery,80,15,6,1200PR07,Snacks,25,40,3,1000PR08,Dairy,55,16,2,880PR09,Fruits,70,22,5,1540PR10,Vegetables,35,28,4,980PR11,Grocery,90,12,6,1080PR12,Snacks,20,45,2,900PR13,Dairy,65,14,3,910PR14,Fruits,75,20,5,1500PR15,Vegetables,45,26,4,1170PR16,Grocery,100,10,7,1000PR17,Snacks,35,32,3,1120PR18,Dairy,50,19,2,950PR19,Fruits,80,18,6,1440PR20,Vegetables,30,35,3,1050"""df = pd.read_csv(StringIO(data))print("First five rows:\n", df.head())print("\nRows and columns:", df.shape)print("\nData types:\n", df.dtypes)print("\nMissing values:\n", df.isnull().sum())print("\nDuplicate rows:", df.duplicated().sum())print("\nStatistical summary:\n", df.describe())

2. Observations

1. The dataset contains 20 products and 6 columns.
2. The total quantity sold is 500 units.
3. Fruits generate the highest category revenue of ₹5,980.
4. Snacks have the highest total quantity sold at 152 units.
5. There are no missing values or duplicate rows.

3. Conclusion

Fruits contribute the highest revenue, while snacks have the highest sales volume. The supermarket can use these patterns to plan inventory and promotions. The provided revenue values equal price multiplied by quantity, without subtracting the listed discounts, so actual discounted revenue should be recalculated if discounts apply.

Question 4: Banking Loan Approval Dataset

1. Python program

import pandas as pdimport matplotlib.pyplot as pltimport seaborn as snsfrom io import StringIOdata = """Customer_ID,Income,Credit_Score,Loan_Amount,Employment_Type,Loan_StatusC01,45000,720,200000,Salaried,ApprovedC02,25000,580,150000,Self-employed,RejectedC03,60000,750,300000,Salaried,ApprovedC04,22000,560,120000,Unemployed,RejectedC05,50000,710,250000,Salaried,ApprovedC06,30000,600,180000,Self-employed,RejectedC07,70000,780,350000,Salaried,ApprovedC08,28000,590,160000,Self-employed,RejectedC09,55000,730,270000,Salaried,ApprovedC10,24000,570,130000,Unemployed,RejectedC11,65000,760,320000,Salaried,ApprovedC12,35000,620,200000,Self-employed,RejectedC13,48000,700,240000,Salaried,ApprovedC14,26000,585,140000,Unemployed,RejectedC15,75000,790,400000,Salaried,ApprovedC16,32000,610,190000,Self-employed,RejectedC17,52000,725,260000,Salaried,ApprovedC18,27000,575,150000,Self-employed,RejectedC19,68000,770,330000,Salaried,ApprovedC20,23000,555,125000,Unemp

loyed,Rejected"""df = pd.read_csv(StringIO(data))print("First five rows:\n", df.head())print("\nRows and columns:", df.shape)print("\nData types:\n", df.dtypes)print("\nMissing values:\n", df.isnull().sum())print("\nDuplicate rows:", df.duplicated().sum())print("\nStatistical summary:\n", df.describe())

2. Observations

1. The dataset contains 20 customers and 6 columns.
2. Ten loans are approved and ten are rejected.
3. All approved customers in this dataset are salaried.
4. Approved customers have higher incomes and credit scores than rejected customers in this sample.
5. No missing values or duplicate rows are present.

3. Conclusion

Income, credit score, and employment type show clear associations with loan approval in the given dataset. Banks can examine these factors during risk assessment, but real loan decisions also require other financial and eligibility details.

Question 5: Agriculture Crop Yield Dataset

1. Python program

import pandas as pdimport matplotlib.pyplot as pltimport seaborn as snsfrom io import StringIOdata = """Farm_ID,Rainfall_mm,Temperature,Fertilizer_kg,Soil_Type,Crop_Yield_kgF01,850,28,60,Loamy,3200F02,600,32,45,Sandy,2100F03,900,27,65,Loamy,3500F04,500,34,40,Sandy,1800F05,780,29,55,Clay,3000F06,650,31,48,Sandy,2300F07,920,26,70,Loamy,3700F08,720,30,52,Clay,2800F09,550,33,42,Sandy,1900F10,800,28,58,Loamy,3100F11,880,27,62,Loamy,3400F12,620,32,46,Sandy,2200F13,760,29,54,Clay,2950F14,940,26,72,Loamy,3800F15,580,33,44,Sandy,2000F16,820,28,60,Clay,3150F17,700,30,50,Clay,2700F18,960,25,75,Loamy,3900F19,540,34,41,Sandy,1850F20,790,29,56,Clay,3050"""df = pd.read_csv(StringIO(data))print("First five rows:\n", df.head())print("\nRows and columns:", df.shape)print("\nData types:\n", df.dtypes)print("\nMissing values:\n", df.isnull().sum())print("\nDuplicate rows:", df.duplicated().sum())print("\nStatistical summary:\n", df.describe())

2. Observations

1. The dataset contains 20 farms and 6 columns.
2. Farms with higher rainfall generally show higher crop yields in this dataset.
3. Farms using more fertilizer generally have higher yields.
4. Loamy soil has the highest average crop yield, followed by clay and sandy soil.
5. No missing values or duplicate rows are present.

3. Conclusion

Rainfall, fertilizer usage, temperature, and soil type are associated with crop yield in this sample. Farmers can use these observations to plan crop production, but more data is needed to confirm the effects of each factor.